

Frequency-Domain Counting & Detection using Fourier Cross-Correlation

by

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Abstract:

This project will develop and evaluate a compact frequency-domain pipeline that detects and counts repeated objects in images (e.g., coins, lemons, or other small, repeated items) by applying Fourier cross-correlation between an image and one or more templates, converting the correlation back to the spatial domain to locate peaks, and adding rotation-invariance via a log-polar transform and phase-only correlation. The work will reproduce the Fourier-based detection recipe from the lab notebooks, quantify counting and localization accuracy (absolute/count error, precision/recall, localization IoU), and analyze robustness to scale, rotation and clutter using a small curated dataset (unsplash examples used in the lab plus 20–50 additional test images). Implementation deliverables will include reproducible Jupyter notebooks and a command-line script, a link to a code repository, and a final written report with visualizations of frequency responses and failure cases; a short demo video may be included.